

# Kanishka Harwani

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## PROFESSIONAL SUMMARY

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Robotics and Mechatronics Engineer with hands-on experience in designing, building, and integrating autonomous mobile robots and embedded systems. Strong expertise in sensor fusion, perception, and real-time control for robotic platforms using ROS 2, micro-ROS, and CAN-based systems. Proficient in C++ and Python, with practical experience on ESP32, Jetson Nano, Raspberry Pi, and custom hardware, supporting autonomous navigation, teleoperation, and indoor mobility research.

## EDUCATION

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| <b>New York University</b> , Brooklyn, NY, USA                                         | 09/2023 – 05/2025 |
| <i>Master of Science, Mechatronics and Robotics</i>                                    | GPA: 3.667        |
| <b>Birla Institute of Technology and Science, Pilani</b> , Hyderabad, Telangana, India | 08/2017 – 06/2021 |
| <i>Bachelor of Engineering, Electrical and Electronics Engineering</i>                 | GPA: 3.0          |

## TECHNICAL SKILLS

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- **Programming Languages:** Python, C, C++ (Embedded Systems)
- **Robotics & Embedded Systems:** Autonomous Navigation, Sensor Fusion, Embedded Control Systems, IoT Systems
- **ML & Computer Vision:** Object Detection, Depth Estimation (Stereo Vision), Feature Extraction, Reinforcement Learning
- **Frameworks & Libraries:** ROS 2, OpenCV, TensorFlow, PyTorch, Scikit-learn, Stable-Baselines3, Gym
- **Hardware & Electronics:** PCB Design & Bring-up, Circuit Simulation (LTspice), Microcontroller Development
- **Design & Simulation Tools:** MATLAB, Simulink, Fusion 360, Eagle CAD, KiCAD
- **Development & OS Tools:** Linux (Ubuntu), Git, Jupyter Notebook, VS Code, CLion, PyCharm, Arduino IDE
- **Communication Protocols:** UART, SPI, I<sup>2</sup>C, CANBUS, Bluetooth, Wi-Fi

## WORK EXPERIENCE

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| <b>Systems Engineer Intern</b> , Gen Auto AI, Piscataway, NJ                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 11/2025 - Present |
| <ul style="list-style-type: none"><li>• Designed and integrated mechanical and electrical systems for an autonomous vehicle test-bed, including sensors, cameras, LiDAR, compute units, and power systems.</li><li>• Developed custom mounting hardware and a drive control board to control steering, throttle, and braking via CAN bus.</li><li>• Collaborated with cross-functional teams to set up remote development environments, provide progress updates, and document results through technical reports and papers.</li></ul> |                   |
| <b>Research Scholar Intern</b> , New York University, MAE Department, Brooklyn, NY                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 09/2025 - 11/2025 |
| <ul style="list-style-type: none"><li>• Designed and prototyped a one-wheel-based reaction-wheel-driven robotic platform, leading mechanical design, electronics integration, and embedded software development.</li><li>• Implemented real-time control and sensor fusion on ESP32 using micro-ROS, IMU, TOF sensors, encoders, and CAN-based motor control via ODrive.</li><li>• Built a modular ROS 2-compatible stack for teleoperation, SLAM, and indoor mobility research, with technical documentation.</li></ul>               |                   |
| <b>Robotics Specialist</b> , Young Tinker Academy, Bhubaneswar, India                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 01/2022 - 08/2022 |
| <ul style="list-style-type: none"><li>• Designed and developed educational robots using Fusion 360, EagleCAD, and TinkerCAD, integrating 3D design, electronics, and programming concepts for student learning.</li><li>• Led a software team to build Bluetooth control apps and a Blockly-based coding platform, while optimizing robot and packaging design to reduce material waste by 18% and production costs by 34%.</li></ul>                                                                                                  |                   |

## PROJECTS

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**ROS2-Enabled Micro Mobility Robot Platform:** ESP32-CAM, C++, Micro-ROS, Dual IMUs, Bluetooth/Xbox Controller

- Built a 4x4-inch ROS2-enabled mobile robot using ESP32-CAM, supporting teleoperation via Bluetooth/Xbox controller and real-time video streaming.
- Integrated dual IMUs, camera, and motor drivers to achieve reliable sensing, control, and indoor mobility.
- Implemented addressable LEDs and buzzer for system feedback, creating a low-cost, modular robotics platform.

**AMR Test Platform:** Jetson Nano, mmWave Radar, ROS2, Sensor Fusion, Python, Dash, Xbox Controller Teleop, ESP32

- Built a four-wheel-drive autonomous mobile robot equipped with mmWave radar, cameras, IMU, and GPS for safety testing in human-robot shared environments.
- Spearheaded the development of a ROS2-based framework for real-time sensor fusion and safety algorithms, enabling comparative testing of radar vs. camera perception and autonomous vs. remote-controlled operation.

**Autonomous Object Retrieval Mobile Robot** OpenCV, Raspberry Pi, Apriltags, Color Detection, Path Planning

- Crafted a vision-guided robot with AprilTag-based localization for autonomous object detection and delivery.
- Designed the robot, electronics, and playing field, and contributed to path planning algorithms on Raspberry Pi.

## PUBLICATION

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“Design and Analysis of Dynamics, Control and Simulation of Y-4 Quadrotor Structure for Hybrid Aerial Vehicle,” IEEE INCET, 2020. DOI: 10.1109/INCET49848.2020.9154033